

ASHESI NEURAL ENGINEERING PROGRAM OUTLINE – FALL 2025

Objectives

The program aims to:

- Provide students with relevant knowledge and skills to pursue further study or work in the field of Neural Engineering (NE).
- Motivate students and spark passion in neural engineering.

After participating in the program, students should, at an introductory level,

- Have a broad understanding of NE and the problems the field addresses.
- Be familiar with current neurotechnology and their applications in NE.
- Be able to record and/or analyze neural /electromyographic signals.
- Have requisite information to seek graduate education in the field.

Course Instructors

- Lee Fisher, PhD - lef44@pitt.edu
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- Robin Lienkämper, PhD - rol96@pitt.edu
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Schedule

Date	Lesson
Sep 30	Mainstream Neurotechnology
Oct 07	Neuro-disorders
Oct 14	Introduction to Neuroscience
Oct 28	Action potential models
Nov 04	Neural Data Recording
Nov 11	Neural Data Analysis
Nov 18	Electromyography
Nov 25	Motor and Sensory Neuroprostheses
Dec 02	Neuro-ethics

Meeting Details

- Time: Tuesdays 17:00 GMT
- Location: Bioengineering Lab, Ashesi